ChiaChan (Henry) Ho

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Education

University of Pennsylvania Expected Jun 2027
Master of Science in Engineering in Systems Engineering

University of California, San Diego Sep 2022 – Jun 2025
Bachelor of Arts in International Business, *Minor in Data Science* GPA: 3.85/4.0
Provost Honors: (Fall 2022 – Spring 2024, 5 quarters)
Relevant Coursework: Data Structures & Algorithms, Probabilistic Modeling & Machine Learning, Linear Algebra

Experience

AI Research Fellow, Handshake AI Solutions LLC – Remote (Philadelphia, PA) Sep 2025 – Present

- Design and author multimodal reasoning tasks (text + images) in the computer science domain, applying reinforcement learning with human feedback (RLHF) to expose weaknesses in LLM reasoning and knowledge retrieval.
- Evaluate AI responses to detect reasoning and retrieval failures, providing structured feedback that informs model refinement and domain adaptation.
- Collaborate asynchronously with peers to ensure task clarity, reproducibility, and adherence to strict quality standards in prompt engineering and data labeling.

Software Engineer Intern, Roshow Group – Zhejiang, China May 2024 – Aug 2024

- Automated daily data cleaning task using Python, reducing manual processing time from 30 minutes to 1 minute to improving team efficiency.
- Improved data processing speed from 2 minutes to 10 seconds by vectorizing Python loop with NumPy in a data transformation pipeline.
- Decreased backend API error rate over 20% by refactoring input validation logic and adding over 50+ unit tests using Pytest.

Data Science Instructional Assistant, Halıcıoğlu Data Science Institute – San Diego, CA Sep 2023 – Aug 2024

- Taught more than 300+ students in introductory and intermediate data science courses, leveraging Python to explain core concepts such as recursion, sorting, and hash maps in discussion sections and office hours.
- Identified common pain points in student code submissions and collaborated with faculty to redesign assignments, reducing conceptual confusion over 20%, as reflected in mid-term score improvements.
- Provided 1-on-1 debugging guidance to more than 100+ students, helping them understand algorithmic thinking and Python debugging workflows using real-world data examples.

Projects

League of Legneds Playing Style Analysis (Link) - San Diego, CA Mar 2024 - Jun 2024

- Engineered a data pipeline in Python using Pandas and NumPy to process 10,000+ professional League of Legends matches; extracted game-level features (e.g., DPM, CKPM) to represent play styles.
- Built and deployed classification models with Scikit-learn to distinguish between LPL and LCK games, obtaining an accuracy rate 70% through feature engineering and hyperparameter tuning.
- Collaborated with project team using Git for version control and reproducible workflows; published results and interactive visualizations on the project website.

Data Science Educational Platform Development (Link) - San Diego, CA Sep 2023 - Dec 2023

- Co-developed an open-source educational site used by 1000+ data science students by developing comprehensive exam solutions and maintaining consistent content structure in Markdown.
- Maintained content accuracy and platform reliability by reviewing peer submissions and student feedback, resolving GitHub issues, and participating in weekly pull requests with faculty advisors.
- Reduced onboarding and editing time for new contributors by standardizing file naming, reorganizing folders, and documenting update workflows across 50+ files.

Technical Skills

Programming Languages: Python, Java, SQL, JavaScript, HTML, MATLAB

Tools/Libraries/Framework: Git, AWS, RLHF, Flask, Spark, boto3, NumPy, Pandas, Scikit-learn, MS Office Suite